Zhalok Rahman

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Github: https://github.com/zhalok
Video Introduction: https://bit.ly/4jpZSJo

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline

Education

Shahjalal University of Science and Technology
Bachelors of Science in Computer Science
2019-01-01 – 2024-04-01

Experience

Kinetik Healthcare *Software Engineer 2*
2025-04-01 – Present

- Built tools for internal system testing, monitoring and tracing
- Improved database performance
- Containerized and deployed applications to **AWS ECS**
- Used **AWS ECR** for managint container images
- Managed AWS infrastructure as code using **CloudFormation**
- Automated deployment with **Codepipeline**
- Configured Route 53 DNS records for routing traffic to cloudfront distribution
- Configured Route 53 DNS records for routing traffic to AWS ALB-backed ECS services
- Managed TLS certificates with **AWS Certificate Manager**
- Deployed static front-end app in **Cloudfront** for edge caching
- Built **Grafana** dashboards for visualizing and exporting metrics, logs and traces from multiple sources
- Set alerts for detecting downtimes and failures upfront
- Investigated incidents and fixed root causes based on **Logs traces and application event**
- Conducted load testing to ensure system reliability at scale before releasing to production
- Writen unit tests to ensure code reliability
- Automated build check, unit test check in continious integration pipeline
- Conducted spec driven development for AI assisted coding
- Managed SDLC with isolated dev, stage and production environments

PropertyFinder *Software Engineer*
2024-10-01 – 2025-10-31

- Maintained reliability and performance of **Golang based miroservices**
- Ingested event logs into **Opensearch** (AWS managed elasticsearch)
- Applied distributed tracing between microservices using **OpenTelemetry**
- Worked as **on-call engineer** to fix production incidents
- **Fixed root cause** of critical production incidents across multiple services using metrics, logs, and traces from **Grafana**
- Conducted **Load Testing** and built dashboards to observe the performance metrics under high stress
- Deployed and managed containerized application in **AWS ECS**
- Managed CI/CD pipeline using **Github Actions**
- Managed infrastructure resources as code with **AWS Cloudformation**
- Followed standard SDLC practices
- Written **unit tests** to ensure code reliability
- Configured alerts in **Grafana** with **Slack** integration for real-time notification on any potential dincident
- Used **Windsurf** for studying legacy codebases efficiently

Learnyx AI *Fullstack Engineer*
2023-10-01 – 2025-02-01

- Integrated **Langfuse** for LLM observability, enabling cost tracking and performance tracing across model calls
- Decoupled heavy workloads from the main request cycle using asynchronous **RabbitMQ** consumers, improving resource efficiency
- Enhanced realtime UX with server side events
- Optimized load performance using a hybrid lazy-loading and preloading strategy
- Acted as a technical advisor in stakeholder discussions, bridging engineering constraints with business goals
- Managed self-hosted containerized infrastructure on **AWS EC2** using **Coolify** and **Docker**, supporting **500+ active users**

Projects

MongoDB vs PostgreSQL benchmarking for storing documents
Benchmark harness comparing MongoDB BSON and PostgreSQL JSONB for document storage under concurrent load, with container and database-internal metrics.

- Built prometheus scrapper for exporting performance metrics and storage footprints for MongoDB and PostgreSQL
- Simulated high concurrency and observed the metrics under high stress
- Benchmark results: https://bit.ly/4dbwS4h
- Github: https://bit.ly/3TYSpGZ

Tech: Golang, JavaScript, Docker, PostgreSQL, MongoDB, Prometheus, Grafana, Loki, Promtail

PostgreSPECTOR
Observability toolkit for PostgreSQL that exposes execution stats and container metrics to Prometheus and Grafana, with logs shipped via Loki.

- Built prometheus scrapper for exporting postgres internal metrics like average query time, CPU usage, heap usage etc.
- Built grafana dashboard for visualizing prometheus metrics
- Simulated stress using pgbench
- Demo: https://youtu.be/w2OCn0X3bRw
- Github: https://bit.ly/4y8AFXX

Tech: Golang, Prometheus, Grafana, PostgreSQL, Loki, Promtail, Docker

Obstainer
Lightweight container observability tool that reads metrics straight from cgroups and exposes them to Prometheus and Grafana.

- Built Golang based prometheus scrapper for capturing container metrics from cgroup
- Extracted docker logs using promtail and pushed to loki
- Built grafana dashboards for visualizing metrics from prometheus and logs from loki data source
- Demo: https://youtu.be/sifxPdFs2AE
- Github: https://bit.ly/4y8AL1L

Tech: Golang, Grafana, Prometheus, Docker, Loki, Promtail

Python TCP Server
Experiments comparing TCP server I/O models (thread-per-connection, non-blocking, event-driven) under load.

- Built simple key-value data store using raw TCP socket using different communication architectures (single threaded, multi-threaded, single threaded event-driven)
- Collected CPU usage, Memory usage, Network usage using cAdvisor
- Benchmark results: https://bit.ly/3V6UUr9
- Github: https://bit.ly/4xMsr7q

Tech: Python, Socket, System Calls, Docker

Certifications

AWS Certified Cloud Practitioner
AWS
https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e
Deep dive on AWS services

Fundamentals of Operating Systems
Udemy
https://www.udemy.com/certificate/UC-ff0d44a3-8917-40cd-aef9-a01e258aee49/
Learned about Process structure, Memory managment, CPU cycles, I/O management

Fundamentals of Network Engineering
Udemy
https://www.udemy.com/certificate/UC-0a49d23e-036a-4c80-ad5a-410ccb877ebb/
Learned about under the hood mechanisms of network protocols